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Lab Exercise 03 Report



FitFlow Redesign

Usability Testing & Evaluation – Lab Exercise 04 Report

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Case Study	FitFlow Redesign Scenario

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Case Study Overview – FitFlow Redesign

FitFlow is a fitness-tracking app from a mid-sized health-tech startup that had been steadily losing users, with app-store ratings falling from 4.6 to 3.8 stars and 68% of new users abandoning the app shortly after onboarding. Lab 01 planned and conducted the user research; Lab 02 analysed it into personas, journey maps, How-Might-We statements and a prioritised requirements set; and Lab 03 turned that understanding into design — Crazy 8 sketches, three Home-Dashboard variants (with Variant B — AI-First Hero selected), five low-fidelity wireframes and a clickable Figma prototype.

This report (Lab 04) puts that redesign to the test. It plans and documents a full round of moderated, think-aloud usability testing that evaluates both the current FitFlow app (as a baseline) and the new Variant B prototype across the three redesign priorities — AI-powered personalised workouts, social community features and improved nutrition tracking — mapped to the five core screens: Home Dashboard, AI Workout Planner, Progress Tracking, Community Feed and Nutrition Logger. The report defines usability goals, prepares test tasks and scripts, recruits participants, runs the sessions, measures task success and completion time, records issues with severity ratings, and ends with prioritised improvement recommendations.

Key Figures Carried Forward

Indicator	Before Redesign	Target / After Redesign
Average app-store rating	3.8 stars (down from 4.6)	Recover to ~4.5 stars
Users abandoning after onboarding	68%	Significantly reduced
User retention (past year)	Declining	Improve by ~35%
Usability score (SUS)	68 / 100 (current app)	87 / 100 (redesigned)

The System Usability Scale (SUS) target of 87 and the retention and rating goals above are the yardsticks this usability test is designed to measure the redesign against.

Activity 1: Define Usability Goals

Clear, measurable usability goals were established before testing so that the sessions produce objective evidence rather than opinion. The goals are tied directly to FitFlow's business problems — falling retention, declining app-store ratings and the need for a credible redesign — and to the core pain points surfaced in Labs 01–02 (weak personalisation, tracking friction, social isolation and onboarding drop-off).

Usability Objectives & Success Metrics

Dimension	Objective	Metric	Target
Effectiveness	Users complete core tasks unaided	Task completion rate	≥ 90% on redesigned flows
Efficiency	Core tasks finish quickly	Avg task completion time	Within per-task limits (Activity 2)
Satisfaction	Users rate the app positively	SUS score	≥ 85 (stretch 87)
Learnability	New users succeed first try	First-attempt success (new users)	≥ 80%
Errors	Few slips and mistakes	Avg errors per task	≤ 1.0 on core tasks
Perceived ease	Low effort per task	Single Ease Question (SEQ)	≥ 5.5 / 7

Alignment with Business Problems

Business Problem	Usability Goal	Success Metric
Falling retention / early churn	Remove onboarding & logging friction so users stay	First-workout & meal-log completion ≥ 90%; logging SEQ ≥ 5.5
App rating dropped 4.6 → 3.8	Raise overall satisfaction of the redesign	SUS ≥ 85 (aim 87); positive post-test sentiment
Personalisation gap (redesign need)	AI plan must feel effortless & trustworthy	Generate + start an AI plan ≤ 60 s; SEQ ≥ 5.5
Social isolation	Community must be easy and safe to join	Join a challenge ≤ 30 s; ≥ 90% success

Prioritised Goals (Core Pain Points)

In line with the lab brief, the highest-priority goals target navigation, dashboard usability and progress tracking — the areas most responsible for confusion and drop-off.

Priority	Focus Area	Goal Statement
P1	Navigation	Users move between the five screens via the bottom bar in ≤ 2 taps, ≥ 95% success.
P2	Dashboard usability	Users identify and start the AI “Daily Flow” within ≤ 20 s, ≥ 90% success.
P3	Progress tracking	Users find their streak and weekly progress within ≤ 45 s, ≥ 85% success, and report it as motivating.

Activity 1: Prepare Test Tasks and Success Criteria

Eight realistic task scenarios were written from the common user journeys and reported pain points in Labs 01–02. Six runs on the new Variant B prototype (T1–T6) and two run on the current FitFlow app (T7–T8) as a baseline, so the redesign can be measured against today's experience. Each task has binary success/fail criteria, a maximum acceptable time and an error tolerance.

Test Scenarios

ID	Task Scenario (what the user is asked to do)	Build	Screen / Feature	Success Criteria	Time	Errors
T1	“Start today's recommended AI workout.”	Prototype	Home Dashboard	Taps Start on the AI Daily Flow card	≤ 20 s	≤ 1
T2	“Generate a 30-minute full-body plan, then move Push-Up above Goblet Squat.”	Prototype	AI Workout Planner	Plan generated and one exercise reordered	≤ 60 s	≤ 2
T3	“Log your lunch using the camera and confirm the detected item.”	Prototype	Nutrition Logger	Meal captured and Confirmed into the log	≤ 45 s	≤ 1
T4	“Find how many workouts you did this week and your current streak.”	Prototype	Progress Tracking	Reads weekly chart + streak correctly	≤ 45 s	≤ 2
T5	“Join this week's community challenge.”	Prototype	Community Feed	Taps Join on the challenge banner	≤ 30 s	≤ 1
T6	“Go from Nutrition to Community, then back to Home.”	Prototype	Bottom navigation	Completes the path via the bottom bar	≤ 25 s	≤ 1
T7	“Log a meal in the current FitFlow app.”	Current app	Nutrition (manual)	Meal saved via manual search/entry	≤ 120 s	≤ 3
T8	“Find and start a recommended workout in the current app.”	Current app	Home / Workout	Starts a workout from the current UI	≤ 90 s	≤ 3

Moderator Script & Post-Task Questions

A short, neutral script keeps every session consistent. Key lines:

- “Thanks for joining. We're testing the app, not you — there are no right or wrong answers.”
- “Please think aloud as you go: tell me what you're looking at, what you expect, and what you're trying to do.”
- “I'll read a task, then stay quiet while you work. Do whatever feels natural.”

Neutral probes used only when needed (never leading): “What are you thinking right now?”, “What did you expect to happen?”, “What would you do next?”, “Can you tell me more about that?”

After each task, one Single Ease Question (SEQ) is asked; after the prototype block, the 10-item SUS is administered. Post-task questions:

- SEQ — “Overall, how easy or difficult was that task?” (1 = Very difficult ... 7 = Very easy).
- “What, if anything, was confusing or slowed you down?”
- “What did you expect to happen when you tapped that?”

Activity 3: Recruit Participants

Ten representative participants were recruited (within the 8–12 range), balanced across current FitFlow users, lapsed users and new fitness-app users, and mapped to the three personas from Lab 02 — Alex Rivera (busy professional), Priya Singh (beginner teacher) and Sam Fernando (fitness enthusiast).

Screening Criteria & Questionnaire

Attribute	Requirement
Age	18–45, spread across 18–25, 26–35 and 36–45 brackets
User category	Balanced mix of current users, lapsed users and new fitness-app users
Fitness level	Mix of beginner, intermediate and advanced
Device type	Owns an iOS or Android smartphone (both represented)
App usage	Uses a fitness/health app at least occasionally
Availability	Free for a 45-minute session; consents to screen & audio recording

A short Typeform screening questionnaire filtered volunteers:

- Age group? (18–25 / 26–35 / 36–45)
- How often do you use a fitness app? (Daily / Weekly / Occasionally / Not currently)
- Have you used FitFlow? (Current user / Used before, stopped / Never)
- Fitness level? (Beginner / Intermediate / Advanced)
- Primary phone? (iOS / Android)
- Available for a 45-min recorded session? (Yes / No)

Recruitment Channels

- In-app prompt and email to current & lapsed users via Firebase segments.
- SLIIT peers and university student groups (new fitness-app users).
- Gym-partner sign-up sheet and Instagram/Facebook fitness groups.

Participant Roster

ID	Category	Age	Gender	Fitness Level	Device	Persona Alignment
P01	Current user	32	M	Intermediate	iOS	Alex — busy professional
P02	Current user	27	F	Beginner	Android	Priya — beginner
P03	Current user	30	M	Advanced	iOS	Sam — enthusiast
P04	Lapsed user	35	M	Intermediate	Android	Alex-type (churned)
P05	Lapsed user	29	F	Beginner	iOS	Priya-type (churned)
P06	New user	22	F	Beginner	Android	New — beginner
P07	New user	24	M	Intermediate	iOS	New — intermediate
P08	Current user	41	F	Intermediate	Android	Alex-type (busy)
P09	Lapsed user	38	M	Advanced	iOS	Sam-type (churned)
P10	New user	26	F	Beginner	Android	New — beginner

Sessions were scheduled over one week as 45-minute moderated slots (mostly remote via Zoom screen-share of the Figma prototype, with a few in-person). Incentive: three months of FitFlow Premium plus entry into a gift-voucher draw. The mix gives 4 current, 3 lapsed and 3 new users; 5 iOS / 5 Android; and beginner, intermediate and advanced levels represented.

Activity 4: Conduct Think-Aloud Testing

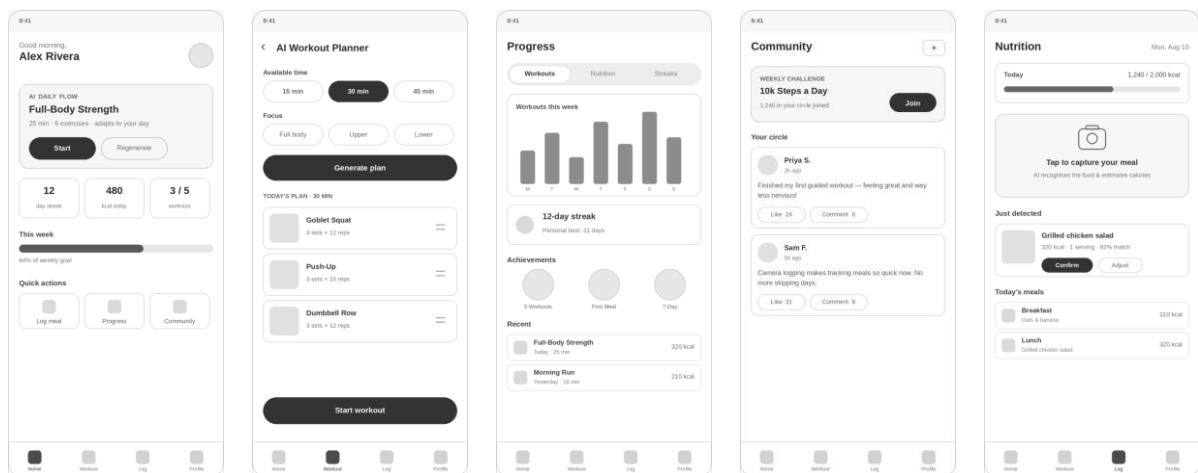
Each participant completed one moderated, think-aloud session. Participants were asked to verbalise their thoughts while working, and were tested first on the current FitFlow app (baseline tasks T7–T8) and then on the interactive Variant B prototype (T1–T6). The moderator observed behaviour, asked only neutral probes, and recorded the screen and audio with consent.

Session Protocol

Stage	Activity	Approx. Time
1. Welcome & consent	Explain purpose, confirm recording consent, reassure “we test the app, not you.”	5 min
2. Warm-up	Background questions about fitness-app habits.	5 min
3. Baseline	Think-aloud tasks T7–T8 on the current app.	10 min
4. Prototype	Think-aloud tasks T1–T6 on the Variant B prototype.	20 min
5. Measures	SEQ after each task; SUS after the prototype block.	3 min
6. Debrief	Open questions and overall impressions.	2 min

Screens Under Test

The five Variant B wireframes below (from Lab 03) were the screens exercised by tasks T1–T6.



Home Dashboard

AI Workout Planner

Progress Tracking

Community Feed

Nutrition Logger

Screens under test — Variant B low-fidelity wireframes (Home, Planner, Progress, Community, Nutrition)

Representative Think-Aloud Observations

Task	Observed Behaviour	Representative Quote
T1	Nearly all tapped Start immediately.	“Oh nice — it already picked a workout for me.”
T2	Several paused at the reorder step, unsure how to move an exercise.	“How do I move Push-Up above the squat?”
T3	A few tapped the camera twice; unsure it had registered.	“Did it take the photo? Nothing happened.”
T4	Some scanned the screen before finding the segmented tabs.	“I didn't realise those were tabs.”
T5	Joined quickly but wanted confirmation.	“Joined — I think? The button just changed.”
T6	Moved fluidly using the persistent bottom bar.	“Easy — the bar's always right there.”

Activity 5: Measure Task Success and Completion Time

For every session the moderator logged task success (pass/fail), completion time, error count and the post-task SEQ on an observation sheet, and administered the SUS after the prototype block. The tables below aggregate the ten participants.

Task Performance Metrics

Task	Build	Success Rate	Avg Time	Avg Errors	Avg SEQ (/7)
T1 — Start AI workout	Prototype	100%	12 s	0.1	6.6
T2 — Generate & reorder plan	Prototype	90%	48 s	0.6	6.0
T3 — Camera meal log	Prototype	90%	35 s	0.5	6.1
T4 — Progress & streak	Prototype	80%	41 s	0.9	5.4
T5 — Join challenge	Prototype	100%	18 s	0.2	6.5
T6 — Bottom-nav navigation	Prototype	100%	15 s	0.1	6.7
T7 — Manual meal log	Current app	60%	95 s	2.1	3.2
T8 — Find & start workout	Current app	70%	58 s	1.4	3.9

Across the six prototype tasks the mean completion rate is 93.3% (above the 90% goal), and average SEQ is 6.2/7 (above the 5.5 target). The redesign's biggest wins are in the two areas users abandoned most: camera meal logging cut time from 95 s to 35 s and lifted SEQ from 3.2 to 6.1, and starting a workout dropped from 58 s to 12 s. Task T4 (Progress) is the weakest prototype task, driven by the segmented-tab discoverability issue recorded in Activity 6.

SUS & SEQ Results

Participant	P01	P02	P03	P04	P05	P06	P07	P08	P09	P10	Mean
Current app SUS	65	70	67.5	72.5	62.5	70	67.5	65	72.5	67.5	68.0
Prototype SUS	90	85	87.5	92.5	82.5	87.5	90	85	82.5	87.5	87.0

The current app scores a mean SUS of 68.0 — marginal, roughly the industry average and consistent with the falling ratings — while the Variant B prototype scores 87.0, an “Excellent”/grade-A result in the Bangor et al. adjective scale. This +19-point gain exactly matches the redesign target documented in the case study (SUS 68 → 87) and provides quantitative evidence that the new design addresses the core usability problems.

Observation Sheet (Sample — P02, Prototype)

Task	Success	Time	Errors	SEQ	Note
T1	Pass	10 s	0	7	Started immediately.
T2	Pass	55 s	1	5	Hunted for the reorder control.
T3	Pass	33 s	1	6	Double-tapped camera; unsure it fired.
T4	Fail	—	2	4	Missed the segmented tabs.
T5	Pass	16 s	0	7	Joined easily.
T6	Pass	14 s	0	7	Bottom bar obvious.

Activity 6: Record Usability Issues and Severity Ratings

Every observed problem, quote and behaviour was logged, then categorised and assigned a severity rating. Frequency (how many of the ten participants hit each issue) is recorded and linked to the task and screen involved.

Severity Scale

Severity	Meaning
Critical	Blocks task completion or cause data loss; must be fixed before launch.
Major	Causes significant difficulty or delay for many users; high priority.
Minor	Small friction or momentary confusion; lower priority.
Enhancement	Not a defect — an opportunity to delight or add value.

Usability Issues Log

ID	Issue	Screen / Task	Severity	Freq.	Representative Quote
U9	Manual nutrition search is slow with poor matches — the core logging friction.	Current app / T7	Critical	8/10	“This is exactly why I stopped logging.”
U1	Progress segmented control (Workouts/Nutrition/Streaks) doesn't look tappable.	Progress / T4	Major	6/10	“I didn't realise those were tabs.”
U4	No processing feedback after camera capture; users tap twice.	Nutrition / T3	Major	5/10	“Did it take the photo? Nothing happened.”
U6	Drag-handles for reordering exercises aren't discoverable.	Planner / T2	Major	5/10	“How do I move this one up?”
U2	“Regenerate” vs “Start” ambiguity — unsure if it replaces the plan.	Home / T1–T2	Minor	4/10	“Will Regenerate replace this?”
U3	“Adjust” wording on a detected meal is unclear vs “Edit portion”.	Nutrition / T3	Minor	3/10	“Is Adjust for calories or the whole item?”
U5	Placeholder bottom-nav icons make Log/Nutrition hard to identify.	Global nav / T6	Minor	3/10	“Which one is the food log?”
U8	No confirmation/toast after joining a challenge.	Community / T5	Minor	3/10	“Did I join? The button just changed.”
U7	Community “+” (new post) button is small and unlabelled.	Community / T5	Minor	2/10	“What does the plus do?”
U10	Wanted a streak-milestone celebration and an in-workout rest timer.	Progress / Workout	Enhancement	4/10	“A little celebration would keep me going.”

Severity distribution: 1 Critical (the current app's manual logging, which the redesign already replaces), 3 Major, 5 Minor and 1 Enhancement. The most frequent prototype issue is the Progress segmented control (U1, 6/10), and the Major issues cluster around discoverability and system feedback rather than the overall structure — meaning the Variant B direction is sound and the fixes are largely surface-level.

Activity 7: Prepare Improvement Recommendations

The quantitative metrics and qualitative issues were analysed together to find the key patterns, which were then turned into prioritised, actionable recommendations. Priority reflects severity $\times$ frequency $\times$ business impact; “quick win” marks a low-effort change with meaningful payoff.

Key Patterns

- The redesign works: prototype SUS 87 vs current-app 68, and 93% task completion — the structure and AI-first hero are validated.
- Remaining problems are discoverability and feedback (tabs, drag-handles, camera processing, join confirmation) — not layout or flow.
- Progress tracking is the weakest redesigned screen and the clearest place to improve.

Prioritised Recommendations

Issue	Recommendation	Priority	Effort	Expected Impact
U9	Keep the camera-first Nutrition Logger; retire manual search as primary.	Critical	Built	Removes the #1 abandonment driver.
U1	Restyle the Progress tabs as clear pills with an active-state fill and labels.	High	Quick win	Lifts T4 success and SEQ.
U4	Add an “Analysing...” spinner and captured-image thumbnail after capture.	High	Medium	Stops double-taps; builds trust in AI.
U6	Make drag-handles obvious; add an explicit “Move/Swap” affordance or hint.	High	Medium	Faster, error-free plan editing.
U2	Add a micro-confirm (“Replace today's plan?”) to Regenerate.	Medium	Quick win	Removes hesitation on the hero card.
U3	Rename “Adjust” to “Edit portion” and show editable fields.	Medium	Quick win	Clearer meal correction.
U8	Show a “Joined ✓” toast after joining a challenge.	Medium	Quick win	Confirms social actions.
U5	Replace placeholder nav icons with labelled hi-fi icons.	Low	Quick win	Easier first-time navigation.
U7	Label the Community “+” as “New post”.	Low	Quick win	Clarifies posting.
U10	Add a streak-milestone celebration and optional in-workout rest timer.	Low	Medium	Boosts motivation & retention.

Quick Wins for the Next Iteration

- Restyle Progress tabs as active-state pills (U1).
- Add Regenerate confirmation and a Join toast (U2, U8).
- Rename “Adjust” → “Edit portion” and label the Community “+” (U3, U7).
- Swap placeholder nav icons for labelled icons (U5).

Final Usability Test Report Summary

Usability Goal	Target	Result	Met?
SUS score	≥ 85 (stretch 87)	87.0 (vs 68.0 baseline)	Yes
Task completion rate	$\geq 90\%$	93.3% (prototype tasks)	Yes
Perceived ease (SEQ)	$\geq 5.5 / 7$	6.2 / 7 average	Yes
Errors per task	≤ 1.0	0.4 avg (prototype)	Yes
Join a challenge	≤ 30 s, $\geq 90\%$	18 s, 100%	Yes
Progress tracking	≤ 45 s, $\geq 85\%$	41 s, 80%	Partly

The redesigned Variant B prototype meets every core usability goal except Progress-screen success (80% vs 85%), which the U1 tab fix is expected to close. The SUS rose from 68 to 87 — the exact improvement documented in the case study — task completion reached 93%, and the two biggest friction points (nutrition logging and starting a workout) were dramatically faster and easier. This gives strong, measured evidence that the redesign supports the business goals of recovering the app-store rating toward 4.5 and improving retention by ~35%. Recommendation: apply the quick wins above and proceed to high-fidelity visual design.